From Global Thresholds to Regional Learning: An Explainable AI Framework for Multi-Hazard Climate Classification in Khuzestan

Mahboubeh Molavi-Arabshahi [a*], Masoumeh Chalangaryamchi [b], Erfan Shahabinezhad [c]

[a] Department of Mathematics & Computer Science, Iran University of Science and Technology, Narmak, Tehran 16846-13114, Iran. E-mail: molavi@iust.ac.ir

[b] Department of Mathematics & Computer Science, Iran University of Science and Technology, Narmak, Tehran 16846-13114, Iran. E-mail: masoumehchalangar@gmail.com

[c] Department of Mathematics & Computer Science, Iran University of Science and Technology, Narmak, Tehran 16846-13114, Iran. E-mail: erfan.shahabi1381@gmail.com

* Corresponding author's E-mail: molavi@iust.ac.ir

Abstract:

Climate hazards increasingly threaten agricultural sustainability in Khuzestan Province, Iran, necessitating reliable and interpretable forecasting tools. This study develops a regionally-calibrated, explainable AI (XAI) framework to classify multiple climate hazards—including drought, heatwaves, and temperature accumulation deficits—across Khuzestan's diverse agro-climatic zones. Using long-term ERA5 reanalysis data (1940–2023), we trained separate XGBoost models for four sub-regions and two critical agricultural seasons (spring and autumn). A key innovation was the strict exclusion of label-defining features during model training to prevent data leakage and ensure generalizability. The models achieved high predictive accuracy, with F1-scores predominantly between 0.80 and 0.90 for most hazards. Predictions for more complex compound events, such as concurrent hot-and-dry conditions, were more challenging, reflecting their inherent physical complexity. SHAP (SHapley Additive exPlanations) analysis provided physically interpretable insights, identifying relative humidity and vapor pressure deficit as the most influential predictors, with notable regional variations in feature importance. This study presents a robust, leakage-aware, and interpretable framework that moves beyond generic thresholds to offer region-specific hazard forecasts. The results demonstrate significant potential for supporting climate-resilient agricultural planning and decision-making in Khuzestan and similar climate-vulnerable regions.

Keywords: Machine Learning, Climate Hazards, SHAP Analysis, ERA5 Reanalysis Data, XGBoost, Khuzestan

INTRODUCTION

Khuzestan Province is one of Iran's most critical agricultural regions, playing a vital role in national food security, rural employment, and livelihoods [1]. The province is a leading producer of wheat, rice, maize, and dates, thanks to its fertile land and water resources from rivers like the Karun and Karkheh. However, the agricultural sector faces severe pressures from declining river discharges and persistent droughts, which directly impact staple crops like wheat and threaten regional food stability [2, 3]. Despite extensive cultivation of crops like date palms, water use

efficiency remains low, highlighting an urgent need for modernized irrigation, farmer training, and updated cropping patterns [4, 5].

In recent years, the intensification of extreme climate events—driven by global warming—has emerged as a major obstacle to sustainable agriculture in Khuzestan [6]. The province increasingly experiences multi-year droughts, intense heatwaves, declining rainfall, and severe fluctuations in soil moisture, patterns consistent with global climate hotspots, including the rapid onset of ecologically damaging 'flash droughts' [7]. Summer temperatures now frequently exceed 50°C, severely impacting crop growth, irrigation efficiency, and livestock health, with experimental studies confirming the adverse effects of heat stress on crop fertility [8]. Consecutive droughts have reduced surface water flows and lowered groundwater levels, while soil over-drying and rising salinity have diminished soil fertility [9]—a challenge long studied through integrated soil-water modeling [10, 11]. Furthermore, climate fluctuations have altered pest and disease cycles, increasing infestation risks and farmers' pest management costs [12]. Collectively, these hazards reduce yield per unit area, lower crop quality, and cause substantial economic losses, potentially undermining rural livelihoods and prompting migration [13]. This escalating threat underscores the critical need for a predictive framework that can detect climate hazards at fine spatial scales and provide interpretable insights to support climate-resilient agricultural planning.

The primary goal of this study is to build a regional climate hazard model to identify and predict climate-vulnerable areas in Khuzestan Province, focusing on hazards such as drought, heatwaves, and precipitation deficits, thereby providing an applicable framework for decision support in agricultural planning [15]. To achieve this, our work is guided by several specific objectives. First, we seek to develop a sub-regionally calibrated, data-driven machine learning framework for multi-hazard climate classification using long-term ERA5 data. Second, we integrate SHAP-based explainability to identify the region-specific drivers of each climate hazard. A third key objective is to implement strict feature exclusion protocols to prevent label-feature leakage and ensure model generalizability. Finally, we comprehensively evaluate model performance across distinct hazard types, sub-regions, and key agricultural seasons.

Previous studies have laid important groundwork, yet critical gaps remain. Many approaches rely on coarse provincial averaging, which obscures critical microclimatic stressors [19], while others lack interpretability, limiting their utility for decision-making [20, 21]. Although machine learning models like XGBoost have proven effective for predicting climatic extremes [23, 25], their application in a leakage-aware, multi-hazard classification framework tailored to sub-regional variability is rare. Furthermore, few studies systematically exclude label-defining features to prevent data leakage or employ post-hoc interpretability methods like SHAP to build trust and provide actionable insights [26].

In this study, we bridge these gaps by developing a leakage-aware, interpretable framework specifically designed for Khuzestan's climatic diversity. We combine the predictive power of

XGBoost [22, 23] with SHAP-driven transparency [24] and strict feature-exclusion protocols. By conducting sub-regional calibration and providing explicit feature attribution, this study establishes a reproducible and interpretable framework for forecasting multi-hazard climate risks across Khuzestan's agriculturally critical zones, offering a robust tool for enhancing climate adaptation and resilience.

2. Data and Methods

2.1. Study Area Description

Khuzestan Province, located in southwestern Iran, is a critically important yet climate-sensitive agricultural region. The province experiences a hot semi-arid to arid climate, characterized by intense summers with frequent heatwaves, droughts, and high radiative loads. This study focused on the land area of Khuzestan, bounded by latitudes 29.5° to 33° N and longitudes 47° to 51° E, with all sea-surface grid points excluded from the analysis to maintain a terrestrial focus.

The region exhibits significant climatic heterogeneity, driven largely by its topography. The northeastern areas adjacent to the Zagros foothills are characterized by higher elevation, which leads to comparatively lower average temperatures and higher humidity levels. In contrast, the southern and southwestern plains experience the highest average temperatures and lowest humidity, creating a distinct and more extreme thermal environment (Fig. 1). This pronounced gradient necessitates a sub-regional approach to accurately capture local hazard dynamics.

To operationalize this, the study domain was divided into four sub-regions: northwest (NW), northeast (NE), southwest (SW), and southeast (SE). The boundaries were placed at 31.25°N and 49°E because these coordinates align with well-documented physiographic and hydroclimatic discontinuities across Khuzestan. The latitude 31.25°N approximates the transition from the cooler, more humid Zagros foothill zone to the markedly hotter and drier southern plains. Likewise, the longitude 49°E corresponds to the principal west–east moisture gradient of the province, separating the relatively humid southwestern lowlands—affected by fluvial and marshland influences—from the drier southeastern interior. Accordingly, these boundary lines follow established topographic and agro-ecological divisions rather than representing arbitrary quadrant cuts. Each terrestrial grid point was programmatically assigned to exactly one sub-region to ensure spatial exclusivity and prevent any overlap between regions.

Given Khuzestan's agricultural calendar, spring and autumn were selected as the primary focus seasons. These periods represent active sowing and harvesting windows, making them critically sensitive to climate hazards, unlike the summer season when many farmlands lie fallow.

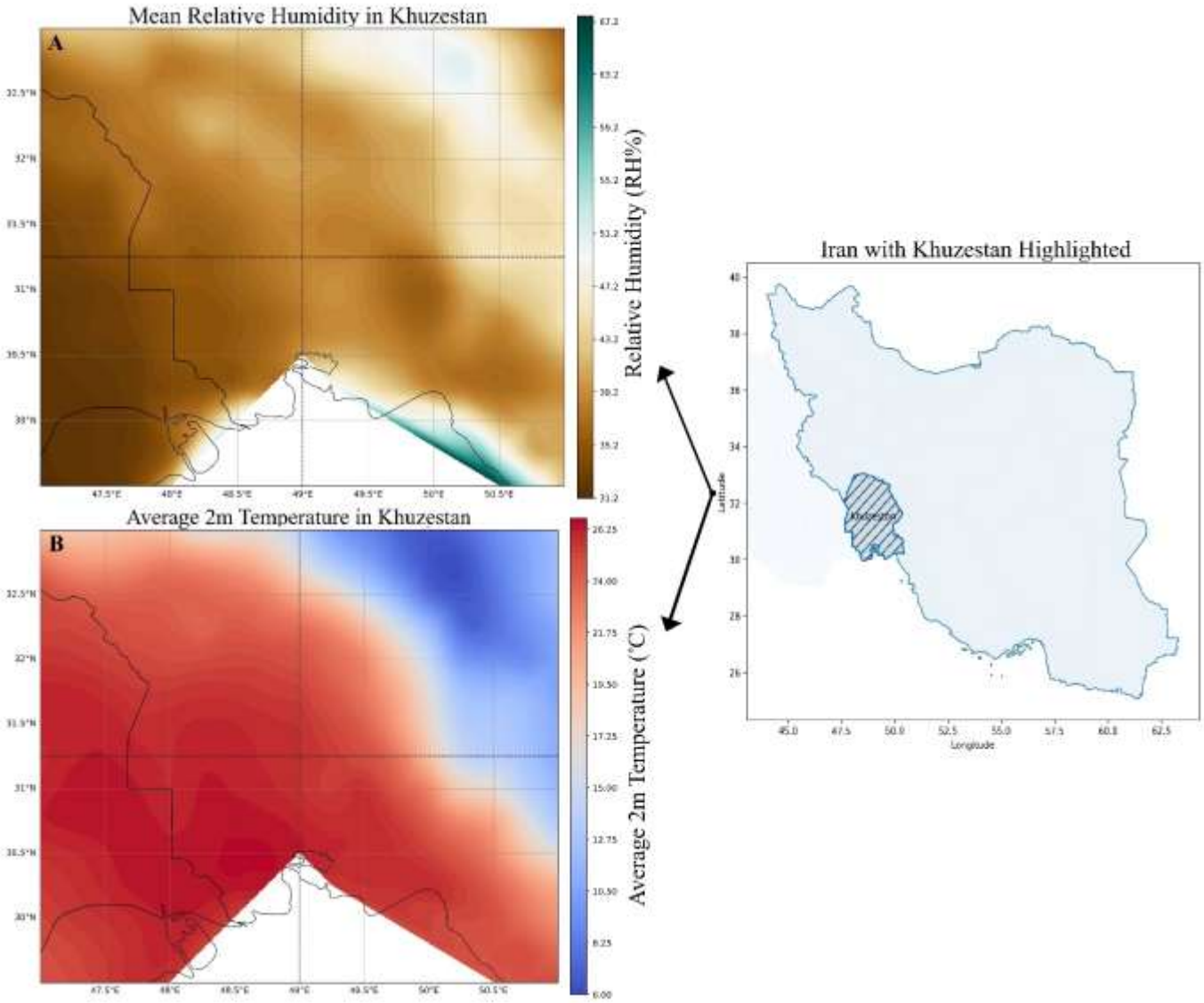


Fig. 1. Long-term Mean Relative Humidity and 2m Air Temperature in Khuzestan. Panel A illustrates the spatial distribution of long-term mean relative humidity across Khuzestan Province. Panel B shows the long-term average 2-meter air temperature (°C). (Data points located over the sea are excluded).

2.2. Data Sources and Variables

The primary source of climate data for this study was the ERA5 reanalysis dataset, provided by the Copernicus Climate Data Store (CDS). The ERA5 Monthly Averaged Data on Single Levels were utilized, which offers a spatial resolution of 0.25 degrees and a long-term record from 1940 onward, making it highly suitable for examining climatic trends and extremes [28, 30]. Data were extracted for the terrestrial area of Khuzestan Province, bounded by latitudes 29.5° to 33°N and longitudes 47° to 51°E. To ensure the analysis remained focused on land-based processes critical for agriculture, all sea-surface grid points were systematically excluded using a world landmask (Natural Earth) in conjunction with spatial libraries like cartopy and shapely.

The initial dataset comprised a set of primary variables directly obtained from the ERA5 archive, which capture key atmospheric, surface, and hydrological conditions. These variables are listed in the table below.

Variable	Description	Units
T2M	2-meter average daily air temperature	°C
D2M	2-meter dew point temperature	°C
TP	Monthly total precipitation	mm
SSRD	Surface incoming shortwave radiation	W/m^2
Z	Geopotential height at 500 hPa	m^2/s^2
SWVL1	Topsoil volumetric moisture (0–7 cm)	%
U10	10-meter eastward wind component	m/s
V10	10-meter northward wind component	m/s
LAI_LV	Leaf Area Index for low vegetation	m^2/m^2
LAI_HV	Leaf Area Index for high vegetation	m^2/m^2

To enrich the feature set and better represent processes relevant to agricultural hazards, several key indices were derived from the primary variables using scientifically validated formulas. Relative Humidity (RH) was calculated from air temperature and dew point to quantify atmospheric moisture content. Vapor Pressure Deficit (VPD), a critical indicator of atmospheric evaporative demand and plant water stress, was derived from RH and saturation vapor pressure. To monitor thermal accumulation essential for crop development, Growing Degree Days (GDD) were computed daily using a base temperature of 10°C, suitable for common summer crops in the region, and then summed monthly. This index provided a reliable representation of crop developmental rates and sensitivity to heat conditions, consistent with established agricultural meteorology literature [31]. Furthermore, the Standardized Precipitation Index (SPI) was employed to quantify precipitation anomalies. This was calculated by fitting a gamma distribution to the seasonal precipitation data and transforming its cumulative distribution function to the standard normal distribution, providing a standardized measure of meteorological drought [32].

The data processing workflow involved multiple stages to prepare the final modeling dataset. After extracting daily and monthly raw ERA5 data, The derived indices (RH, VPD, GDD) were calculated. Anomalies for all variables were then computed relative to the long-term means over

the study period to isolate deviations from normal conditions. The data were subsequently aggregated into seasonal summaries for spring and autumn, applying appropriate operations such as calculating the mean for temperature variables or summing for precipitation and GDD. This processed dataset was structured at the grid-cell level for each year, season, and climatic sub-region, preserving spatial resolution while capturing seasonal climate variability across Khuzestan Province from 1940 through 2023.

2.3 Data Preprocessing and Modeling Approach

2.3.1 Feature Engineering and Hazard Label Construction

Before training the machine learning models, it was essential to transform raw climatic observations into meaningful and standardized variables that can effectively represent the underlying environmental processes. This step, known as feature engineering, played a crucial role in ensuring that the model can capture the physical and statistical relationships between atmospheric drivers and hazard occurrences. In parallel, the definition of hazard labels determines how climatic extremes are categorized and learned by the model. Together, these two components formed the foundation of the modeling framework by converting the raw ERA5 reanalysis data into a structured, machine-readable form suitable for predictive analysis. The following section described the procedures adopted for feature computation, normalization, and the construction of regionally calibrated hazard labels used in model training.

In this study, reanalysis data from the ERA5 dataset were utilized to construct the model's training dataset. The data included selected atmospheric, soil, and climatic variables collected for two seasons—spring and autumn—across four climatic sub-regions of Khuzestan Province (northwest, northeast, southwest, and southeast). Spring and autumn represent active periods for sowing and harvesting, making them more critical for hazard monitoring. Furthermore, as illustrated, the clear variability in model performance across both regions and seasons confirmed the appropriateness of treating sub-regions and seasons separately. This approach effectively captures the distinct climatic behaviors and hazard dynamics specific to each spatial and temporal context (Fig. 2).

Data preprocessing steps ensure clean, standardized, and leakage-free inputs. Following the loading of regional datasets, a preprocessing stage was carried out, including the removal of missing values. Subsequently, the data were split into training and testing subsets with a 70:30 ratio. Feature normalization using StandardScaler was then applied by fitting the scaler only on the training data and transforming both the training and testing sets accordingly. This measure significantly reduced the likelihood of data leakage from the test set into the training process. In addition to the raw variables, several derived features such as Growing Degree Days (GDD), Vapor Pressure Deficit (VPD), and Relative Humidity (RH) were calculated using established formulas based on surface air temperature, dew point, and surface pressure. These computations enhanced the climatic feature set and contributed significantly to improving the model's predictive performance [33].

For the initial labeling process, threshold-based criteria grounded in internationally accepted hazard definitions were applied. For example, the drought hazard was identified through a composite condition involving four indicators: the Standardized Precipitation Index (SPI),

temperature anomalies (t2m_anom), surface soil moisture anomalies (swvl1_anom), and vapor pressure deficit anomalies (VPD_anom). After filtering the dataset for missing values, these multivariate thresholds were used to assign binary labels (hazard presence or absence). To further calibrate these thresholds and better align them with the regional climatic patterns of Khuzestan, a post-labeling validation step using Principal Component Analysis (PCA) was implemented. PCA was applied to the normalized climatic variables (excluding the hazard label itself) to project the data into a two-dimensional space. By visualizing the distribution of labeled samples within this PCA space, the separability of classes was assessed [34]. No automated threshold optimization algorithm was applied. Instead, PCA visualization was used as a supportive diagnostic tool to check whether the initially selected thresholds behaved consistently across regions. Any adjustments were made only to ensure coherence with regional climatic patterns and were confirmed through subsequent model performance checks. This step helped verify that the threshold-based labels remained stable and regionally appropriate without relying on data-driven optimization.

It is worth noting that unsupervised clustering methods were not considered appropriate for this task, as climatic hazards such as drought or heatwaves are defined operationally through threshold exceedances rather than emerging from natural statistical groupings. Since unsupervised algorithms primarily form clusters based on geometric similarity in the feature space, their outputs do not necessarily correspond to physically meaningful hazard categories or agricultural impact criteria. For this reason, a threshold-based strategy was adopted from the outset to maintain interpretability and consistency with established hazard-monitoring practices.

Although threshold refinement in this study was performed through visual assessment of class separability in the two-dimensional PCA space, the authors acknowledge the inherently subjective nature of this approach. The primary objective of this step was to achieve better alignment between hazard labels and region-specific climatic patterns. To enhance objectivity and reproducibility, future implementations could incorporate quantitative clustering quality metrics—such as the Silhouette Score or the Davies-Bouldin Index—to validate threshold selection and guide data-driven refinement of thresholds. It is worth noting that the refinements introduced at this stage were indirectly validated through improved classification performance and greater consistency in SHAP-based explanatory results during subsequent modeling phases.

Strict feature exclusion was implemented to maintain a leakage-aware modeling framework. To strictly prevent data leakage and ensure the integrity of the learning process, for each hazard model, the most important variables used directly in that specific hazard's label definition (e.g., SPI for drought, t2m anomaly for heatwave) were excluded from the input feature set. However, those variables could still be included in models for other hazards where they were not part of the labeling logic. This separation ensured that the classifier could not simply memorize the labeling logic but instead had to generalize from correlated but independent variables, preserving both statistical validity and generalizability. In this study, this design choice forms the basis of what we refer to as a leakage-aware framework, meaning that the modeling pipeline is explicitly constructed to prevent data leakage—i.e., the unintended use of variables that directly encode the hazard label. By ensuring that label-defining variables are excluded during training, the framework compels the model to learn genuine climatic drivers rather than implicitly reproducing threshold rules.

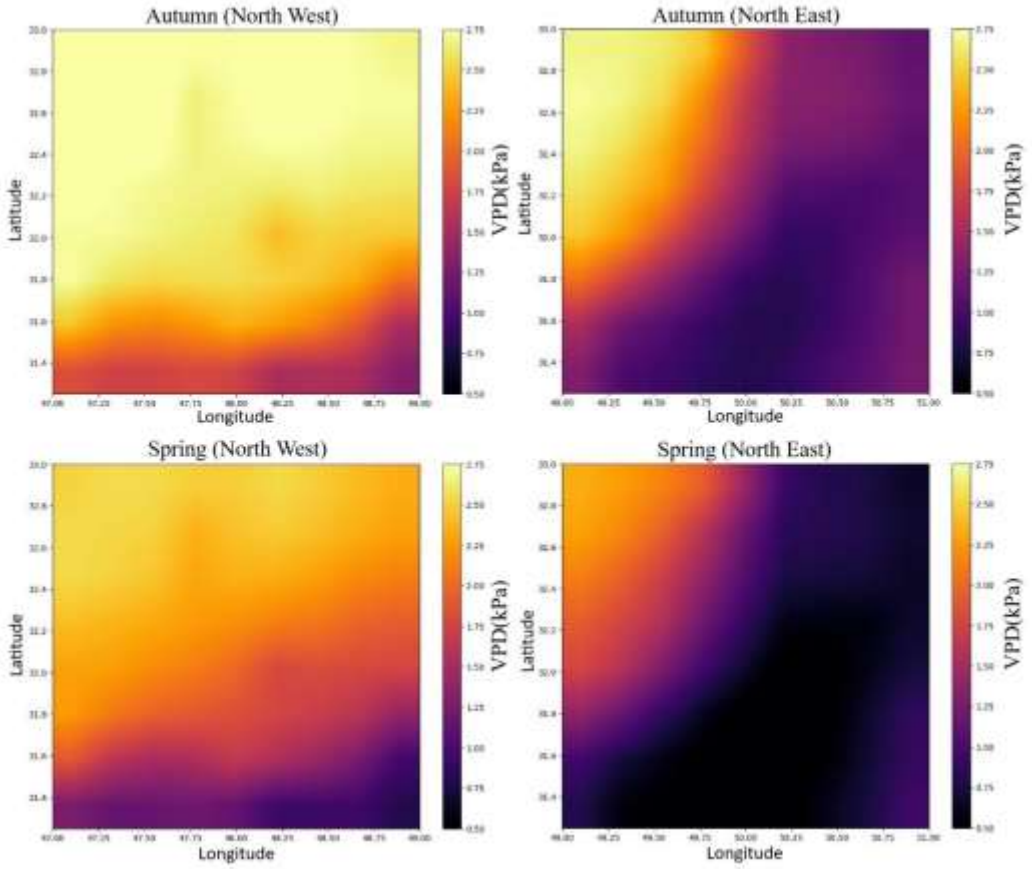


Fig. 2. Seasonal and Regional Variability of Vapor Pressure Deficit (VPD) Across Two Sub-Regions of Khuzestan, both for Autumn and Spring. Higher VPD values in the northwestern sub-region reflect greater atmospheric dryness and evaporative demand, while lower VPD in the northeastern zone corresponds to more humid, topographically influenced conditions during both seasons.

2.3.2 Machine Learning Model Development and Training

This subsection outlined the core workflow for developing machine learning models for hazard prediction. After the construction of features and hazard labels described in the previous section, the next step involves developing predictive models capable of identifying and classifying climatic hazards across Khuzestan Province. This stage represents the core analytical component of the framework, where the structured data are used to train machine learning algorithms to recognize complex nonlinear relationships among climatic variables. The objective of this subsection was to describe the model development workflow, including algorithm selection, parameter optimization, and performance evaluation. A particular focus is placed on the XGBoost algorithm due to its strong capability in handling high-dimensional data, managing imbalanced classes, and providing explainable outputs through SHAP-based interpretability.

Model training followed a structured procedure involving data splitting, hyperparameter tuning, and performance assessment. The modeling was conducted using the XGBoost algorithm. A separate model was developed for each hazard type across each sub-region and season. The dataset was initially split into training and testing sets, followed by hyperparameter tuning through grid search and cross-validation for each model configuration. To evaluate model performance, several

statistical metrics were used, including the F1-score, balanced accuracy**,** area under the ROC curve (AUC), and the confusion matrix. Based on the analysis, the models performed highly accurately to predict some of the hazards, like temperature accumulation, which follow more predictable climatic patterns. Lastly, feature importance analysis was done through the utilization of SHAP values.

Feature importance patterns highlighted key climatic variables contributing to hazard prediction. Similar machine learning approaches have been applied to environmental quality evaluation using multisource remote sensing data [35]. Findings indicated surface air temperature (t2m), total precipitation (tp), and soil moisture indices (swvl) as key variables in the prediction of hazards in various models. Interestingly, the Standardized Precipitation Index (SPI) also served as a robust predictor in some models where it was not utilized in hazard labeling. For example, whereas SPI was omitted from the drought model to prevent label-feature overlap, it was incorporated and was a very key component of models for other hazards like radiation deficit or heatwave.

A schematic flowchart summarizes the full methodological sequence from data acquisition to model interpretability. The methodological workflow of this study is illustrated in a schematic flowchart, outlining the sequential processes from ERA5 data acquisition and regional aggregation to hazard labeling, model training, and interpretability assessment using SHAP and performance metrics (Fig. 3). This is an easy illustration of the process, and it enables readers to understand the sequence of steps, their relationship with each other, and the rationale behind the methodological technique.

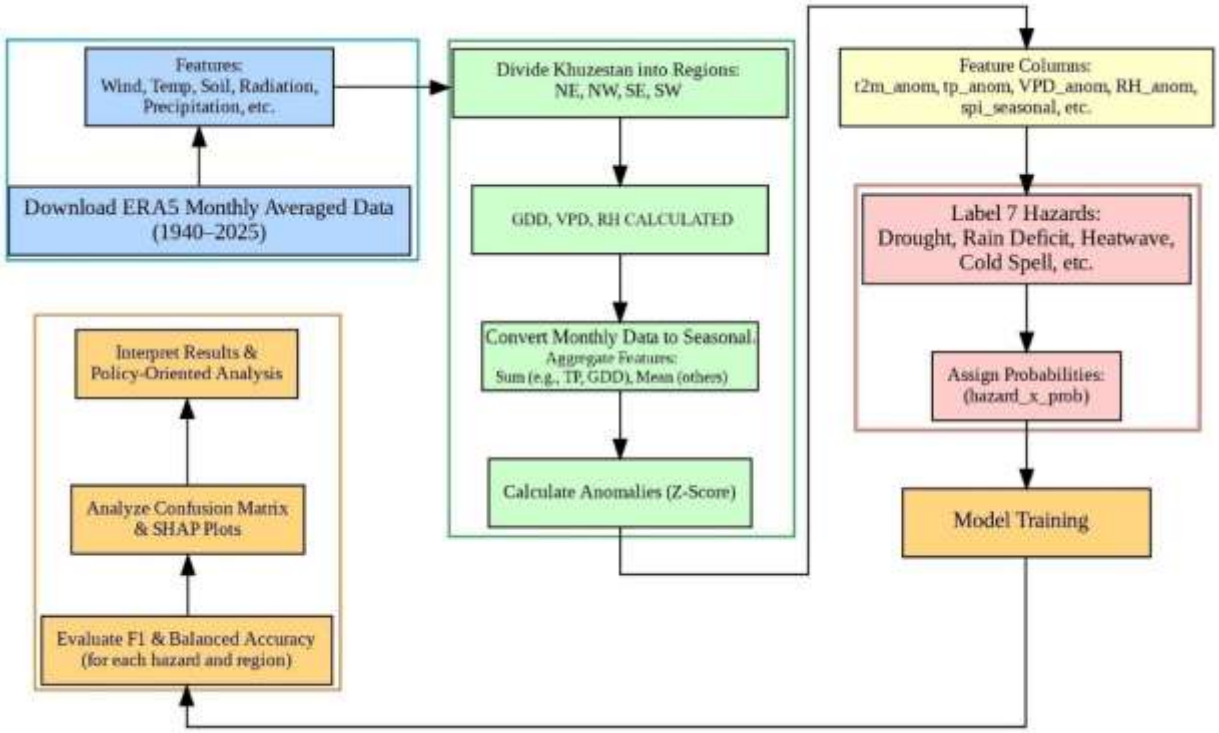


Fig. 3. Schematic representation of the methodological framework adopted in this study, showing steps from ERA5 data processing and regional subdivision to hazard labeling, model training, interpretability analysis, and illustrating how these components together form a coherent, leakage-aware workflow.

3. Results and Model Interpretation

3.1 Classification Accuracy Across Hazards and Sub-Regions

This subsection evaluates the predictive reliability of the models across hazards, regions, and seasons. To assess the reliability and predictive strength of the developed models, classification accuracy was first evaluated across different hazard types, seasons, and climatic sub-regions of Khuzestan. This analysis provides a comprehensive overview of how the framework performs under varying climatic conditions and for both frequent and rare hazard events. By examining the model's ability to correctly classify the presence or absence of specific hazards, this subsection establishes the empirical foundation for subsequent interpretation and spatial analysis.

Overall model performance was strong across most hazards, particularly in regions with consistent climatic patterns. The models showed consistent performance in forecasting climate-related hazards during spring and autumn across the four agro-climatic zones of Khuzestan Province. The machine learning framework based on the XGBoost algorithm, combined with optimization strategies such as class weighting, regularization, purposeful exclusion of label-influencing features, and ٥-fold cross-validation, led to models that achieved a desirable balance between accuracy and generalizability. Among the hazards, models for heatwaves, temperature-accumulation deficits, and radiation deficits performed particularly well—especially in the